

Hooks Events Reference

Day 2 Handout - All 27 Events & 4 Handler Types (July 2026, v2.1.141+)

Hooks are **hard enforcement**: external handlers that run on lifecycle events. Skills advise Claude; hooks don't ask. Use hooks for anything compliance-grade - secret blocking, audit logging, formatting, policy gates.

The 27 Events, Grouped

1. Session Lifecycle (6)

Event	Fires when
SessionStart	A session begins
Setup	Initial environment setup
SessionEnd	Session terminates
InstructionsLoaded	CLAUDE.md/memory loaded
ConfigChange	Settings change mid-session
CwdChanged	Working directory changes

Example - session audit (command handler):

```
{ "hooks": { "SessionStart": [ { "hooks": [
{ "type": "command", "command": ".claude/hooks/log-session.sh" } ] } ] } }
```

```
log-session.sh: echo "$(date -Is) session started in $PWD" >> ~/.claude-sessions.log
```

2. Prompt & Response Flow (7)

Event	Fires when
UserPromptSubmit	You submit a prompt (can inject context or block)
UserPromptExpansion	Prompt expansion occurs
Stop	Claude finishes responding
StopFailure	A stop fails
Notification	Claude Code raises a notification
Elicitation / ElicitationResult	Interactive input requested / answered

Example - prompt guard (prompt handler): on UserPromptSubmit, a `"type": "prompt"` handler evaluates the prompt against policy ("block anything referencing production credentials") before Claude sees it.

3. Tool Execution (6) - the workhorses

Event	Fires when
PreToolUse	Before a tool runs - CAN BLOCK (permissionDecision deny)
PostToolUse	After a tool succeeds - react/format/log
PostToolUseFailure	After a tool fails
PostToolBatch	After a parallel tool batch
PermissionRequest	A permission prompt is shown
PermissionDenied	A permission is denied

Example - the Lab 2B pair (command handlers):

```
{ "hooks": {
  "PreToolUse": [ { "matcher": "Write",
    "hooks": [ { "type": "command", "command": ".claude/hooks/pre-write-check.sh" } ] } ],
  "PostToolUse": [ { "matcher": "Write|Edit",
    "hooks": [ { "type": "command", "command": ".claude/hooks/post-edit-format.sh" } ] } ]
}
```

Block from PreToolUse by printing:

```
{"hookSpecificOutput":{"hookEventName":"PreToolUse","permissionDecision":"deny","permissionDecisionReason":"Hardcoded secret detected"}}
```

Exit 0 with no output = allow.

4. Agents & Tasks (5)

Event	Fires when
SubagentStart / SubagentStop	A subagent spawns / completes
TeammateIdle	An agent-team teammate goes idle (Teams is experimental - verify at delivery - fast-moving)
TaskCreated / TaskCompleted	A tracked task is created / completed

Example - delegation telemetry (HTTP handler): on SubagentStop, "type": "http" posts {agent, duration, tokens} to your metrics endpoint - multi-agent cost visibility for free.

5. Workspace & Context (5)

Event	Fires when
FileChanged	A watched file changes
WorktreeCreate / WorktreeRemove	Worktree lifecycle (parallel sessions, teams)
PreCompact / PostCompact	Around context compaction

Example - compaction safety net (MCP tool handler): on PreCompact, "type": "mcp_tool" calls your notes MCP server to snapshot open TODOs before the window compresses.

The 4 Handler Types

Type	Runs	Use for
command	A shell script (JSON on stdin)	Formatters, blockers, loggers - most hooks
http	POST to an endpoint	Telemetry, ChatOps, central policy services
mcp_tool	An MCP server tool	Reuse existing integrations as handlers
prompt	An LLM evaluation	Judgment calls that regex can't express

Rules That Bite

1. **Matchers are exact + case-sensitive:** `Write`, `Edit`, `Bash` - never lowercase. "`Write|Edit`" matches both.
2. **Reload to activate:** `/clear` (terminal) or restart (VS Code) after editing hooks.
3. **Exit codes matter:** command handlers - exit 0 allows; deny JSON on stdout blocks (`PreToolUse`).
4. **Keep handlers fast:** the July 2026 `/doctor` flags slow hooks; a 2-second hook on `PostToolUse` taxes every edit.
5. **Settings merge:** project `.claude/settings.json` + plugin `hooks/hooks.json` + enterprise `managed-settings.json` (locked) all contribute.

Choosing an Event - Quick Decisions

You want to...	Event	Handler
Block secrets before they hit disk	<code>PreToolUse (Write)</code>	command
Auto-format after edits	<code>PostToolUse</code> , matcher <code>Write+Edit</code>	command
Audit-log every tool call	<code>PostToolUse (*)</code>	command
Alert the team when a long run finishes	<code>Stop</code>	http
Track subagent costs	<code>SubagentStop</code>	http
Preserve state before compaction	<code>PreCompact</code>	mcp_tool or command
Enforce prompt policy	<code>UserPromptSubmit</code>	prompt

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